

Engineering Mathematics III

ENGE702

Lecture 5: Solving linear first order differential equations

Linear first order ODEs

A first order ODE is said to be **linear** if it can be written in the form

$$y' + p(x)y = r(x).$$

For many differential equations of this type, we can solve using a technique involving what is called an **integrating factor**.

For example, the following first order ODEs are linear:

▶ $y' + \sin(x)y = 2x$

▶ $y' + 2xy - \cos(x) = 3$

▶ $e^x y' - 2 \ln(x)y + 5x = 2x^3$

Linear first order ODEs

Before we start to apply the integrating factor technique, it is best to put it in standard form, ie $y' + p(x)y = r(x)$. If there is a term in front of y' , we can simply divide the whole equation by that term.

Now we need to find what is called the integrating factor, defined as

$$\mu(x) = e^{\int p(x) dx}$$

This integrating factor is designed specifically such that when you multiply it through the original differential equation it becomes:

$$(\mu(x)y)' = \mu(x)r(x).$$

Note you can set the arbitrary constant in the result for $\int p(x) dx$ to be zero, since a non-zero value simply results in multiplying the whole equation by a constant which may be cancelled.

We can then integrate both sides and solve for y to get:

$$y = \frac{1}{\mu(x)} \left(\int \mu(x)r(x) dx \right)$$

Linear first order ODEs

We have then solved the differential equation - as long as we can calculate $\int \mu(x)f(x) dx$. This is not always so easy, but for many differential equations it will be possible.

Example: Solve the differential equation

$$y' + 2y = e^{-4x+7}$$

We have: $p(x) = 2$, $r(x) = e^{-4x+7}$, $\mu(x) = e^{2x}$.

So,

$$y = e^{-2x} \int (e^{2x} e^{-4x+7} dx)$$

$$y = e^{-2x} \int e^{-2x+7} dx$$

$$y = -\frac{1}{2} e^{-2x} e^{-2x+7} + c e^{-2x}$$

$$y = -\frac{1}{2} e^{-4x+7} + c e^{-2x}$$

Linear first order ODEs

Example: Solve the initial value problem

$$x^2 y' + 2xy = 3x + 1, \quad y(1) = 1$$

This is not in standard form, so we convert it to standard form by dividing through by x^2 to get:

$$y' + \frac{2}{x}y = \frac{3}{x} + \frac{1}{x^2}$$

We have: $p(x) = \frac{2}{x}$, $r(x) = \frac{3}{x} + \frac{1}{x^2}$, $\mu(x) = e^{\int \frac{2}{x} dx} = e^{2 \ln(x)} = x^2$.

So

$$y = x^{-2} \left(\int \left(x^2 \left(\frac{3}{x} + \frac{1}{x^2} \right) \right) dx \right)$$

$$y = x^{-2} \left(\int 3x + 1 dx \right)$$

$$y = x^{-2} \left(\frac{3x^2}{2} + x \right) + cx^{-2}$$

$$y = \frac{3}{2} + \frac{1}{x} + \frac{c}{x^2}$$

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Now we apply the initial condition: $y(1) = 1$, so:

$$1 = \frac{3}{2} + \frac{1}{1} + \frac{c}{1^2}$$

$$1 = \frac{3}{2} + 1 + c$$

$$c = -\frac{3}{2}$$

Therefore, the particular solution is

$$y = \frac{3}{2} + \frac{1}{x} - \frac{3}{2x^2}$$

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Example: An electrical circuit has an EMF of $E = 48V$ (constant), a resistor of $R = 11\Omega$ and an inductor of $L = 0.1H$. The current is initially 0. Find the current in the circuit as a function of time.

To solve this we need some laws of physics. The first is Ohm's Law: a current in the circuit causes a voltage drop across the resistor of IR , so $V_1 = IR$.

The second is Kirchoff's Law: a current in the circuit causes a voltage drop across the inductor of $LI' = L\frac{dI}{dt}$, so $V_2 = LI'$

Finally, the sum of these two voltage drops gives the EMF, so $E = V_1 + V_2$, or

$$LI' + RI = E.$$

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We will solve this differential equation. We will keep the numbers given as letters for now - this won't make it much harder to do, and may come in handy later if we want to change things.

First we need to put the equation in standard form:

$$I' + \frac{R}{L}I = \frac{E}{L}.$$

Here we have: $p(t) = \frac{R}{L}$, $r(t) = \frac{E}{L}$, and $\mu(t) = e^{\int \frac{R}{L} dt} = e^{\frac{R}{L}t}$.

So,

$$I = e^{-\frac{R}{L}t} \left(\int e^{\frac{R}{L}t} \frac{E}{L} dt \right) + ce^{-\frac{R}{L}t}$$

$$I = e^{-\frac{R}{L}t} \left(\frac{E}{L} \frac{L}{R} e^{\frac{R}{L}t} \right) + ce^{-\frac{R}{L}t}$$

$$I = \frac{E}{R} + ce^{-\frac{R}{L}t}$$

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Inputting our values for E , R , and L , we have

$$I = \frac{48}{11} + ce^{-110t}$$

Now we can put in the initial condition $I(0) = 0$, and get

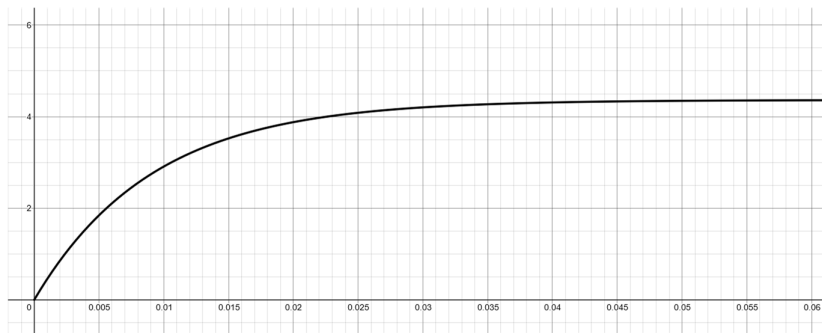
$$0 = \frac{48}{11} + ce^0$$

$$c = -\frac{48}{11}$$

so our current over time becomes

$$I = \frac{48}{11} (1 - e^{-110t})$$

Linear first order ODEs



Exercises

- ▶ Solve the following ODEs:

- ▶ $y' + xy = y$

- ▶ $y' + \frac{3x}{2}y = 2x$

- ▶ A bath is run to a temperature of 40° . The temperature in the bathroom is 18° . After 15 minutes the bath-water is 36° . How long before the water reaches 33° ?

Text Book References

Chapter 1.5